Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers

From Finite Screen Capacity to Local Coupling, Capacity-Flow Normal Forms, and Integrable Geometry Closure

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Abstract

Finite Distinction Systems — Gravity 1 (FDS-G1) Core consolidates the finite screen-capacity gravity program into a single flagship paper. This flagship paper, together with six technical companions (Companions A–F covering screen statistics, transition dynamics, scalar-capacity EFT, likelihood tests, horizon occupancy residuals, and holographic dictionaries), forms the complete FDS-G1 program. The primitive object is a finite causal-screen entropy functional

$$S_{\rm scr} : \mathcal{X}_{\rm phys} \to \mathbb{R},$$

defined on a physical quotient state space of screen variables. Its response one-form

$$\omega = \eta_i(X)\,\mathrm{d}X^i, \qquad \eta_i = \frac{1}{k_{\rm B}}\frac{\delta S_{\rm scr}}{\delta X^i}$$

contains the local area response

$$\eta_{\rm loc}(x) = \omega(\delta_A(x)) = \frac{1}{k_{\rm B}}\frac{\delta S_{\rm scr}}{\delta A(x)}.$$

Under the standard horizon-entropy and local-screen thermodynamic bridges, this area response determines the local inverse coupling, or gravitational stiffness:

$$G^{-1}_{\rm loc}(x) = \frac{4\hbar}{c^3}\eta_{\rm loc}(x), \qquad G_{\rm loc}(x) = \frac{c^3}{4\hbar}[\eta_{\rm loc}(x)]^{-1}.$$

The paper organizes the G1 program around the chain

$$\text{finite screen statistics} \to S_{\rm scr}[X] \to \omega \to \eta_{\rm loc} \to G_{\rm loc} \to \theta\text{-flow} \to \text{integrable geometry} \to \mathcal{R}_{\mu\nu}.$$

Finite screen statistics supplies the area-extensive substrate; interacting screens renormalize the response density; stochastic entropy-gradient screen dynamics gives Raychaudhuri-like capacity-flow normal forms; integrable all-null response data generate Einstein-like metric closure; non-integrable response data appear as residual sectors. Scalar-capacity EFT, holographic dictionaries, horizon thermodynamics, and likelihood-ready observational tests are treated as companion interfaces or manuscripts. The paper is a bridge program: the derivation is conditional on explicitly stated assumptions. The framework is weakened if finite screen statistics fails to produce an area-extensive response substrate, if local response data cannot be metric-integrated, or if residual sectors fail conservation and likelihood penalties. It aims to provide a precise statistical and response-geometric substrate for the area-entropy coefficient that controls gravity-like coupling. The choice of $S_{\rm scr}$ is motivated by a primitive FDS argument: finite maintainable distinctions require causal boundary accounting, and the coordinate-free ledger of finite screen state-count is entropy. A stronger calibration route is developed in D0, where the usual horizon-entropy bridge is replaced by a conditional modular anomaly-inflow argument: the primitive coefficient is a finite-access capacity density η_0; under boost modular calibration and all-null Ward/Bianchi closure one obtains $G = c^3/(4\hbar\eta_0)$ and $S_H/k_B = \eta_0 A_H = A_H/(4\ell_P^2)$.

1 Scope, Claim Status, and Bridge Assumptions

This paper presents the core formulation of the FDS-G1 finite screen-capacity program. The framework is organized around the response-geometric chain

$$\text{finite screen statistics} \to S_{\mathrm{scr}}[X] \to \omega \to \eta_{\mathrm{loc}} \to G_{\mathrm{loc}} \to \theta\text{-flow} \to \text{integrable metric closure} \to \mathcal{R}_{\mu\nu}.$$

FDS-G1 is not introduced as a phenomenological modification of the Friedmann equation or of the Einstein equation. It is a finite-screen response program in which local inverse coupling, capacity-flow normal forms, metric-integrability conditions, and residual sectors arise from a common entropy-response architecture.

1.1 Why finite screen entropy is the primitive object

The choice of a finite causal-screen entropy functional is not an ornamental assumption added after the fact. It is the minimal gravitational primitive obtained by applying the FDS ontology to a causal spacetime setting. The argument is programmatic rather than a derivation of the Einstein equation:

1. Physical distinctions must be finite and maintainable.
2. Maintainable distinctions require a boundary.
3. In gravity, the relevant boundary is causal: an operational screen or horizon.
4. A finite causal screen has a finite state-count ledger.
5. The coordinate-free ledger of finite state-count is entropy.
6. Therefore the primitive gravitational object is a screen entropy functional

$$S_{\mathrm{scr}} : \mathcal{X}_{\mathrm{phys}} \to \mathbb{R}.$$

7. Geometry is the integrable response envelope of this ledger.

Thus, within FDS, finite screen entropy is not a decorative analogy. It is what finite, boundary-maintained distinguishability becomes when the physical access relation is gravitational and causal. Coupling, capacity flow, metric closure, and dark residual sectors are then read as response, integrability, and failure modes of this entropy ledger.

Equivalently, to reject $S_{\mathrm{scr}}[\mathcal{X}_{\mathrm{phys}}]$ as the primitive starting point for the G1 branch, one must reject at least one of the following FDS commitments: finite distinguishability, boundary-maintained access, causal-screen localization, entropy as a coordinate-free capacity ledger, or geometry-as-response.

The central claim is conditional but sharply defined. Given the standard horizon-entropy normalization and local-screen thermodynamic bridge, FDS-G1 identifies the local inverse gravitational coupling, or stiffness, with the local entropy-area response of a finite causal screen:

$$\boxed{G_{\mathrm{loc}}^{-1}(x) = \frac{4\hbar}{c^3}\,\frac{1}{k_{\mathrm{B}}}\,\frac{\delta S_{\mathrm{scr}}}{\delta A(x)}, \qquad G_{\mathrm{loc}}(x) = \frac{c^3}{4\hbar}\left[\frac{1}{k_{\mathrm{B}}}\,\frac{\delta S_{\mathrm{scr}}}{\delta A(x)}\right]^{-1}.}$$

The purpose of this paper is not to replace the Bekenstein-Hawking entropy formula, Jacobson-type local thermodynamic arguments, Raychaudhuri focusing, Lovelock uniqueness, spin-2 consistency, scalar-tensor EFT, or holographic duality. Rather, it supplies a finite screen-capacity substrate and response-geometric language for the area-entropy coefficient that appears in these standard structures.

Claims in this paper are classified as derived toy results, response principles, bridge assumptions, EFT embeddings, normal-form results, or falsification interfaces. This classification distinguishes what follows internally from finite-screen assumptions, what is imported from standard horizon or holographic physics, and what remains an open closure condition.

Central slogan. A useful but nonliteral slogan is:

Gravity-like geometry is the integrable hydrodynamic response of finite causal-screen entropy.

The technical reading is that local screen entropy response defines inverse coupling (stiffness); stochastic screen dynamics defines capacity flow; and metric geometry is the integrable envelope of compatible local response laws.

What would count as a stronger derivation? To upgrade the program from bridge to derivation, one would need to derive internally, from FDS primitives alone, the Bekenstein-Hawking coefficient $1/(4\ell_P^2)$, the local screen first law, Raychaudhuri focusing, all-null tensor closure, and the observed gravitational constants. G1 does not yet do this. It instead provides a sequence of toy theorems, response principles, EFT embeddings, and falsification interfaces. D0 supplies a stronger conditional calibration route: the Bekenstein–Hawking coefficient problem is reduced to two sharper lemmas—construct a causal finite-index screen-algebra net with universal local capacity density η_0, and prove boost modular calibration $K_{\mathcal{S}} = (2\pi/\hbar)Q_\chi$ for local causal screens; once these hold, $G = c^3/(4\hbar\eta_0)$ and $S_H/k_B = A_H/(4\ell_P^2)$ follow.

The comparative claim is structural. A competing macro-spacetime residual theory is not equivalent merely because it can fit a distance curve or introduce effective functions. It must provide a comparable primitive layer, a comparable response chain, and a comparable falsifiable residual fingerprint.

2 Prior-art boundary

The paper uses established results as bridges. Area entropy and black-hole thermodynamics trace to Bekenstein and Hawking [1, 2]. The local thermodynamic route from horizon first law to Einstein equations follows Jacobson-type reasoning [4]. Raychaudhuri focusing is standard congruence geometry [3]. Lovelock uniqueness constrains low-derivative metric closure [5]; spin-2 consistency routes are associated with Weinberg and Deser [6, 7]. Scalar-tensor and Horndeski theories provide the conservative EFT home for variable-coupling sectors [8, 9]. AdS/CFT and holographic entropy are represented by Maldacena, Ryu-Takayanagi, and FLM [10, 11, 12]. GW170817 motivates luminal tensor-sector choices in late-time modified gravity [13].

FDS-G1's contribution is orthogonal to these results: it supplies the finite screen-capacity interpretation and the entropy-response formalism that organizes them into a single bridge program.

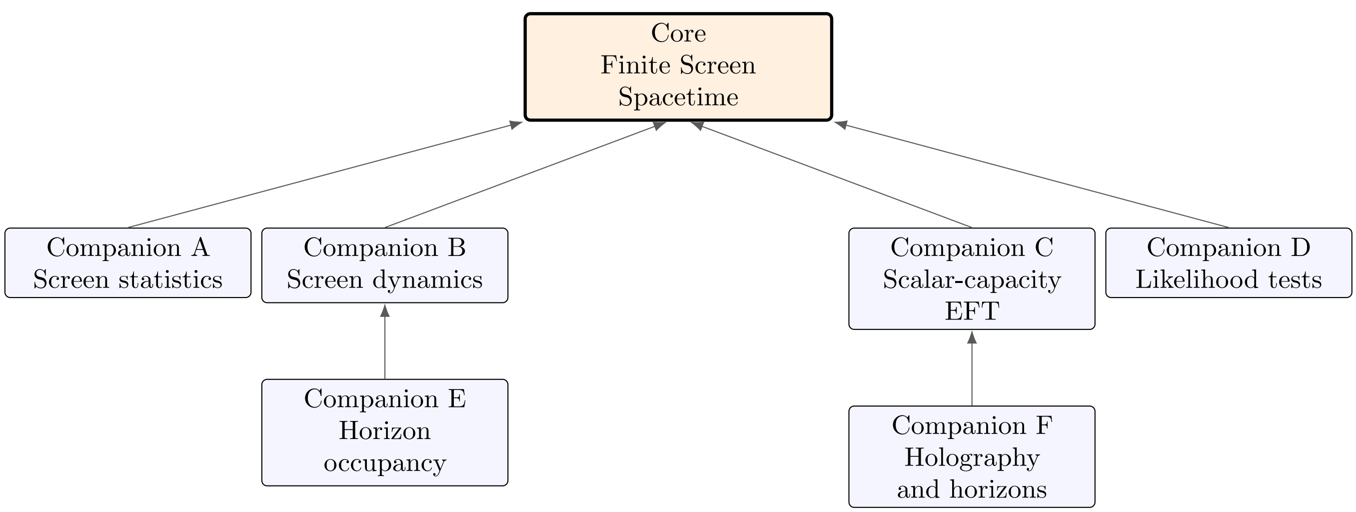
Antiba's finite-capacity thermodynamics of causal horizons studies horizon capacity as a limitation on local classical representability [16]. G1 is adjacent but distinct: it treats finite causal-screen entropy response as the source of local inverse coupling and as input data for metric-integrability and residual-closure structure. Bousso's covariant entropy conjecture links light-sheet area to entropy content [17] and provides a key prior constraint for any causal-screen entropy program. The holographic dictionary section draws on the Brown–Henneaux central-charge relation $c_{\mathrm{CFT}} = 3L/(2G_3)$ as a concrete example of screen capacity density controlling asymptotic coupling [18]. The horizon thermodynamics and first-law discussion complements the Bekenstein–Hawking and Jacobson references with the Iyer–Wald Noether-charge formalism for diffeomorphism-invariant theories [19]. The local thermodynamic route to gravitational equations is further supported by Jacobson's entanglement-equilibrium approach to the Einstein equation [20].

Recent adjacent work has also moved toward finite-capacity, finite-resolution, and information-ledger descriptions of gravitational screens. Barzi et al. study finite heat-capacity corrections to Unruh/Jacobson thermodynamics and the associated Rényi-entropy structure [21]. Nusbaumer formulates semiclassical gravity on causal diamonds with finite-resolution boundary algebras and a relative-entropy variational principle [22]. Song develops a local entropy-balance ledger on causal horizons, balancing area entropy, modular energy, and irreversible record updates [23]. Ahmad and Klinger develop an algebraic quantum-probability route to emergent area terms and generalized entropy [24]. Bianconi formulates an entropic action from quantum relative entropy between spacetime and matter-induced metrics [25]. These approaches are adjacent to G1 but differ in emphasis: G1 identifies the local inverse coupling with the marginal entropy-area response of a finite causal screen and then follows that response through capacity-flow, metric-integrability, residual closure, and likelihood-ready tests.

Relation to the FDS series. FDS-G1 is the gravitational branch of the broader Finite Distinction Systems program. The formal FDS core defines active finite distinction systems as finite-capacity boundary-maintaining systems and introduces the proof-status discipline used here [28]. FDS-T1 develops observer-relative distinguishability budgets [29]; FDS-T2 formulates effective geometry as horizon boundary accounting [30]; FDS-T3 studies capacity overflow, effective stochasticity, and Markov closure under finite projection [31]; and FDS-X1 gives the predecessor horizon-ledger dark-energy bridge [32]. FDS-X3 decomposes the four known interactions into functional distinction-operation classes, identifying gravity as global causal-geometric / stress-energy accounting [34]. FDS-G1 develops this gravitational row by modeling global boundary accounting as finite causal-screen entropy response. FDS-X2 supplies the weak-sector orientation-capacity module used by X3, but is not a direct input to the gravitational response bridge developed here [33]. The present paper differs from those predecessors by promoting the screen entropy functional $S_{\mathrm{scr}}[X]$, its response one-form ω, and the local area response η_{loc} into the central gravity-specific architecture.

3 Program architecture and dependency graph

The G1 program is organized as one flagship core paper together with six technical companions:



Internal theoretical chain:

finite screen statistics $\rightarrow$ $S_{\mathrm{scr}}[X]$ $\rightarrow$ ω $\rightarrow$ η_{loc} $\rightarrow$ $G_{\mathrm{loc}}^{-1}/G_{\mathrm{loc}}$ $\rightarrow$ θ-flow $\rightarrow$ integrable metric closure $\rightarrow$ $\mathcal{R}_{\mu\nu}$.

Table 1: Claim-status audit for FDS-G1 Core.

Claim	Meaning	Status
Finite cell area law	$S_A/k_\mathrm{B} = (A/a_0)\ln m$	toy theorem
Interacting area law	$S_A/k_\mathrm{B} = \eta_\mathrm{eff} A + o(A)$	statistical bridge
Screen quotient state space	$\mathcal{X}_\mathrm{phys} = \mathcal{X}_\mathrm{raw}/\mathcal{G}_\mathrm{scr}$	definition/framework
FDS primitive screen ledger	finite maintainable distinctions require causal boundary entropy ledger	programmatic / ontological bridge
Response one-form	$\omega = \eta_i \mathrm{d}X^i$	definition
Local coupling bridge	$G_\mathrm{loc} = (c^3/4\hbar)[\eta_\mathrm{loc}]^{-1}$	conditional bridge
Capacity-flow normal form	$\dot{\theta} = -a\theta^2 - b\sigma^2 - c\Phi + \Xi$	stochastic normal-form result
Metric all-null closure	null response reconstructs tensor up to trace	standard tensor bridge
Ward/Bianchi closure	response conservation	gauge-to-Ward bridge
Scalar-capacity EFT	$\varphi = \ln(\eta/\eta_0)$	conservative scalar-tensor EFT
Residual taxonomy	non-integrable response classes	framework
Likelihood interface	SN/BAO/growth/local-G blocks	medium-prior evidence executed; top-control wide-prior sensitivity completed; production refinement pending

4 Finite screen statistical mechanics

Definition 1 (Finite distinction screen)**.** *A finite distinction screen Σ_A is a causal boundary patch with operational area A and finite screen-local distinguishability units. In the simplest model,*

$$N_A = \frac{A}{a_0}, \tag{1}$$

where a_0 is an effective area per independent cell.

If each cell carries m distinguishable states and the cells are independent,

$$\Omega_A = m^{N_A}, \qquad S_A = k_\mathrm{B} \ln \Omega_A = k_\mathrm{B} \frac{A}{a_0} \ln m. \tag{2}$$

Therefore

$$\frac{S_A}{k_\mathrm{B}} = \eta_\mathrm{nat} A, \qquad \eta_\mathrm{nat} = \frac{\ln m}{a_0}. \tag{3}$$

Matching the standard horizon bridge gives

$$\eta_\mathrm{BH} = \frac{1}{4\ell_P^2} = \frac{c^3}{4G\hbar}, \qquad G = \frac{c^3}{4\hbar}[\eta_\mathrm{BH}]^{-1}. \tag{4}$$

This identifies the screen-capacity density that must be supplied to reproduce the standard horizon coefficient; the coefficient 1/4 itself remains an external input.

Definition 2 (Screen area-law universality class)**.** *$\mathcal{U}_\mathrm{scr}$ is the class of finite screen systems whose degrees of freedom are screen-local, finite-state, finite-range or sufficiently short-range, finite-correlation in the regime considered, stable under screen-local coarse-graining, and not dominated by hidden bulk-volume degrees of freedom or global nonlocal constraints.*

For systems in $\mathcal{U}_{\rm scr}$ one expects the thermodynamic form

$$\frac{S_A}{k_{\rm B}} = \eta_{\rm eff} A + S_{\rm sub}(A), \qquad \frac{S_{\rm sub}(A)}{A} \to 0 \quad (A \to \infty). \tag{5}$$

Finite interactions renormalize $\eta_{\rm eff}$ rather than destroying the leading area law. Criticality, long-range constraints, infinite local Hilbert spaces, hidden bulk volume degrees of freedom, or dominant nonlocal constraints are controlled failure modes.

5 Screen state space and quotient structure

A screen entropy functional must be defined on physical screen states, not on arbitrary coordinate labels. Write raw variables as

$$X_{\rm raw} = \{A(x), q_{AB}(x), \chi(x), \Phi(x), M(x), \ldots\}, \tag{6}$$

where A is area or cell density, q_{AB} is screen shape, χ is occupancy or residual capacity, Φ is flux load, and M denotes memory or hidden screen modes. The physical state is the quotient

$$\boxed{\mathcal{X}_{\rm phys} = \mathcal{X}_{\rm raw}/\mathcal{G}_{\rm scr},} \tag{7}$$

where $\mathcal{G}_{\rm scr}$ includes screen coordinate reparametrizations, null-generator reparametrizations, cell relabelings, and internal labels that do not change causal access or entropy response.

Definition 3 (Screen entropy functional)**.** *The entropy functional is*

$$S_{\rm scr} : \mathcal{X}_{\rm phys} \to \mathbb{R}. \tag{8}$$

Functional derivatives are understood on the quotient tangent space, or after gauge fixing and projection away from pure gauge directions.

Example 1 (Cell relabeling)**.** *If cells $i = 1, \ldots, N$ have states σ_i, a permutation $i \mapsto \pi(i)$ changes labels but not area, entropy, or response. Thus $\{\sigma_i\} \sim \{\sigma_{\pi(i)}\}$.*

Example 2 (Generator reparametrization)**.** *A monotonic change $\lambda \mapsto f(\lambda)$ changes the coordinate expression for $\theta = {\rm d}\ln A/{\rm d}\lambda$. It does not change the cross-sectional entropy-area response $\eta_{\rm loc} = k_{\rm B}^{-1}\delta S/\delta A$.*

6 Response one-form and local coupling

Let $s_{\rm scr} = S_{\rm scr}/k_{\rm B}$. The response one-form is

$$\omega = \eta_i(X)\, {\rm d}X^i, \qquad \eta_i(X) = \frac{\delta s_{\rm scr}}{\delta X^i}. \tag{9}$$

For field variables this denotes

$$\omega = \int_\Sigma {\rm d}^{d-2}x\, \eta_i(x)\delta X^i(x). \tag{10}$$

Let $\delta_A(x)$ be the physical tangent vector that changes the local area element $A(x)$ while holding the other physical screen variables fixed. Then

$$\boxed{\eta_{\rm loc}(x) = \omega(\delta_A(x)) = \frac{1}{k_{\rm B}}\frac{\delta S_{\rm scr}}{\delta A(x)}.} \tag{11}$$

This makes $\eta_{\rm loc}$ the area component of the full response one-form.

Principle 1 (Local response coupling)**.** *Under the standard horizon-entropy and local-screen thermodynamic bridges, the local area response determines the local inverse coupling, or gravitational stiffness:*

$$\boxed{G_{\mathrm{loc}}^{-1}(x) = \frac{4\hbar}{c^3}\eta_{\mathrm{loc}}(x), \qquad G_{\mathrm{loc}}(x) = \frac{c^3}{4\hbar}[\eta_{\mathrm{loc}}(x)]^{-1}.} \tag{12}$$

For homogeneous screens,

$$\eta_{\mathrm{loc}} \to \eta_\delta(A) = \frac{\mathrm{d}(S/k_{\mathrm{B}})}{\mathrm{d}A}, \qquad G_\delta(A) = \frac{c^3}{4\hbar}[\eta_\delta(A)]^{-1}, \qquad G_\delta^{-1}(A) = \frac{4\hbar}{c^3}\eta_\delta(A). \tag{13}$$

For finite-area corrections

$$\frac{S(A)}{k_{\mathrm{B}}} = \eta_0 A + \alpha \ln(A/A_0) + \gamma + b(A/A_0)^{-p} + \cdots, \tag{14}$$

one obtains

$$\eta_\delta(A) = \eta_0 + \frac{\alpha}{A} - \frac{pb}{A_0}(A/A_0)^{-p-1} + \cdots. \tag{15}$$

Thus macroscopic finite-area corrections are naturally suppressed by inverse area unless a separate residual or scalar-capacity mode is activated.

7 Entropy-potential integrability

If the response one-form derives from a scalar entropy potential s_{scr}, then

$$\omega = \mathrm{d}s_{\mathrm{scr}}, \qquad \boxed{\mathrm{d}\omega = 0.} \tag{16}$$

In coordinates,

$$\mathrm{d}\omega = \frac{1}{2}(\partial_i \eta_j - \partial_j \eta_i)\mathrm{d}X^i \wedge \mathrm{d}X^j. \tag{17}$$

A nonzero response curl indicates missing variables, memory, nonlocal constraints, stochastic projection, or physical residual modes.

Example 3 (Two-variable exact response)**.** *Let*

$$s(A,\chi) = \eta_0 A + \alpha \ln A + \beta A\chi - \frac{\lambda}{2}\chi^2. \tag{18}$$

Then

$$\eta_A = \frac{\partial s}{\partial A} = \eta_0 + \frac{\alpha}{A} + \beta\chi, \qquad \eta_\chi = \frac{\partial s}{\partial \chi} = \beta A - \lambda\chi. \tag{19}$$

The response one-form $\omega = \eta_A \mathrm{d}A + \eta_\chi \mathrm{d}\chi$ is exact, since

$$\partial_\chi \eta_A = \beta = \partial_A \eta_\chi, \qquad \mathrm{d}\omega = 0. \tag{20}$$

Example 4 (Apparent curl from hidden memory)**.** *Suppose the true entropy depends on a hidden memory mode M,*

$$s(A,\chi,M) = \eta_0 A + \alpha \ln A + \beta A\chi + \gamma AM - \frac{\lambda}{2}\chi^2 - \frac{\mu}{2}M^2. \tag{21}$$

On the full space (A,χ,M), $\omega = \mathrm{d}s$ is closed. If M is projected out but depends on history, the reduced apparent response $\tilde{\omega} = \tilde{\eta}_A \mathrm{d}A + \tilde{\eta}_\chi \mathrm{d}\chi$ can satisfy $\mathrm{d}\tilde{\omega} \neq 0$. This is an epistemic or memory residual, not necessarily new fundamental physics.

8 Response Hessian, fluctuations, and scalar-capacity modes

The entropy-response Hessian is

$$\mathcal{H}_{ij} = -\frac{\delta^2 s_{\rm scr}}{\delta X^i \delta X^j}. \tag{22}$$

Near equilibrium, $\mathcal{H}$ controls stability, fluctuations, and slow hydrodynamic response modes. In a Gaussian approximation,

$$\langle \delta X^i \delta X^j \rangle \sim (\mathcal{H}^{-1})^{ij}. \tag{23}$$

The scalar-capacity variable

$$\varphi(x) = \ln \frac{\eta(x)}{\eta_0} \tag{24}$$

can be interpreted as a long-wavelength mode of the area-response density. Its EFT potential is a coarse-grained free-energy cost for moving away from the equilibrium response density. Thus the scalar-capacity EFT is not introduced as a generic dark scalar; it is the hydrodynamic EFT of a slow screen-response mode.

9 Stochastic screen dynamics and capacity-flow normal forms

Let screen macrostate variables $X^i(\lambda)$ evolve by an entropy-gradient stochastic process. A generic Onsager form is

$$\dot{X}^i = L^{ij} \frac{\delta s_{\rm scr}}{\delta X^j} + \zeta^i, \tag{25}$$

with positive semidefinite response matrix L^{ij} and noise ζ^i. A large-deviation or Onsager-Machlup path weight may be written schematically as

$$P[X(\lambda)] \sim \exp[-N_A \mathcal{I}_{\rm scr}[X]], \tag{26}$$

$$\mathcal{I}_{\rm scr}[X] = \frac{1}{2} \int \mathrm{d}\lambda \, (\dot{X}^i - F^i[X]) D^{-1}_{ij} (\dot{X}^j - F^j[X]), \tag{27}$$

where $F^i = L^{ij} \delta s / \delta X^j$ in local equilibrium.

Define

$$N(\lambda) = \frac{A(\lambda)}{a_0}, \qquad x(\lambda) = \ln N(\lambda), \qquad \theta = \dot{x} = \frac{\mathrm{d} \ln A}{\mathrm{d}\lambda}. \tag{28}$$

Then

$$\dot{\theta} = \frac{\ddot{A}}{A} - \left(\frac{\dot{A}}{A} \right)^2. \tag{29}$$

The negative quadratic term is therefore kinematic: it follows from logarithmic capacity expansion. Shape modes contribute the positive scalar $\sigma^2 = \sigma_{AB}\sigma^{AB}$; positive flux load focuses the screen. The low-order affine equilibrium normal form is

$$\boxed{\dot{\theta} = -a\theta^2 - b\sigma^2 - c\Phi_{\rm flux} + \Xi_{\rm res} + \zeta_\theta + \mathcal{O}(3).} \tag{30}$$

Equation (30) is a normal-form bridge: it shows that Raychaudhuri-like terms are stable low-order structures in stochastic finite screen dynamics. The identification of $\Phi_{\rm flux}$ with $R_{\mu\nu}k^\mu k^\nu$ or $T_{\mu\nu}k^\mu k^\nu$ is the standard local-screen geometry bridge.

10 Metric integrability and all-null closure

Entropy integrability is not yet metric integrability. For each event p and null direction k^μ, local screen response data may define a null focusing response $\mathcal{F}_p(k)$.

Proposition 1 (All-null metric envelope)**.** *Let screen response data define a null focusing function $\mathcal{F}_p(k)$ at each event p and null direction k^μ. Assume:*

1. *locality: $\mathcal{F}_p(k)$ depends only on data at p;*
2. *smoothness in p;*
3. *homogeneous quadratic dependence on k;*
4. *compatibility with a Lorentzian null cone;*
5. *nondegenerate screen response in at least two independent null directions.*

Then there exists a symmetric tensor $E_{\mu\nu}(p)$ such that

$$\mathcal{F}_p(k) = E_{\mu\nu}(p)k^\mu k^\nu \tag{31}$$

up to a trace term $\Phi g_{\mu\nu}$ that is fixed by Ward/Bianchi closure.

All-null data reconstructs $E_{\mu\nu}$ up to a trace term, since $g_{\mu\nu}k^\mu k^\nu = 0$. Thus if

$$E_{\mu\nu}k^\mu k^\nu = \kappa T_{\mu\nu}k^\mu k^\nu \qquad \forall k^\mu \text{ null}, \tag{32}$$

then

$$E_{\mu\nu} - \kappa T_{\mu\nu} = \Phi g_{\mu\nu}. \tag{33}$$

Ward/Bianchi closure fixes the trace freedom into the usual cosmological integration sector. In the Einstein-like equilibrium case,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} + \mathcal{R}_{\mu\nu}. \tag{34}$$

Here $\mathcal{R}_{\mu\nu}$ is not an arbitrary dark term; it represents non-integrable or unmodeled screen response data after attempting metric closure.

11 Ward/Bianchi dynamical closure

If the screen response action or its hydrodynamic envelope is invariant under screen reparametrizations that become spacetime diffeomorphism redundancy in the metric envelope, then the effective geometric tensor obeys a Ward identity

$$\nabla^\mu E_{\mu\nu} = 0. \tag{35}$$

Therefore residual sectors must satisfy

$$\boxed{\nabla^\mu \left(T_{\mu\nu} + \mathcal{R}_{\mu\nu}\right) = 0.} \tag{36}$$

This is the conservation-closure condition that prevents variable capacity from becoming an unconstrained varying-G ansatz.

If memory is present, local response may become nonlocal:

$$\eta_i(\lambda) = \int^\lambda \mathrm{d}\lambda'\, K_{ij}(\lambda,\lambda')X^j(\lambda'), \tag{37}$$

or covariantly,

$$\mathcal{R}^{\mathrm{mem}}_{\mu\nu}(x) = \int \mathrm{d}^4x' \sqrt{-g(x')}\, K_{\mu\nu}{}^{\alpha\beta}(x,x')\, T_{\alpha\beta}(x'). \tag{38}$$

Such terms must still obey the total conservation law (36).

12 Residual taxonomy

The residual sector is classified by origin rather than used as a catch-all.

Table 2: Residual taxonomy and G1 companion branches.

Residual type	Meaning	G1 branch
Epistemic	missing coarse variable or projection artifact	model refinement / response
Capacity	$\eta_{\rm loc}$ varies as a slow physical mode	scalar-capacity EFT
Homogeneous	background horizon occupancy pressure	G1DE
Stochastic	finite-cell noise in screen updates	G1micro
Memory	non-Markovian output response; source–kernel factorization is a realization layer	G1response / G1 memory quotient / D6
Lensing/growth	null response differs from matter response	G1obs / G1pred / G1fit
Holographic	bulk/boundary entropy correction	G1Holography
Geometric	no smooth metric envelope exists	G1response core

This classification makes explicit what survives when a residual model fails. For example, if a horizon occupancy model is ruled out, the finite screen statistics and entropy-response framework may survive. If metric closure fails, the program moves beyond standard metric geometry and must state so directly.

13 Scalar-capacity EFT interface

The local response density may have a long-wavelength mode

$$\varphi = \ln \frac{\eta}{\eta_0}, \qquad \frac{G_{\rm eff}}{G_0} = e^{-\varphi}. \tag{39}$$

A conservative scalar-capacity EFT is

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{2} F(\varphi) R - \frac{1}{2} Z(\varphi)(\nabla\varphi)^2 - V(\varphi) + \mathcal{L}_m + \mathcal{L}_{\rm res} \right]. \tag{40}$$

With $F(\varphi) = M_{*,0}^2 e^{\varphi}$, this is a scalar-tensor/Horndeski-compatible sector, not a new field-equation form. Its novelty is the screen-capacity interpretation and prior structure. A minimal Horndeski embedding is

$$G_2 = Z(\varphi)X - V(\varphi), \qquad G_3 = 0, \qquad G_4 = \frac{1}{2}F(\varphi), \qquad G_5 = 0. \tag{41}$$

In EFT-of-dark-energy language,

$$M_*^2 = F, \qquad \alpha_M = \frac{d \ln F}{d \ln a}, \qquad \alpha_T = 0, \qquad \alpha_B = -\alpha_M \tag{42}$$

for the minimal nonminimally coupled model. Stability requires $F > 0$, positive kinetic normalization, absence of gradient instabilities, and compatibility with local-gravity and gravitational-wave propagation constraints.

14 Observational signatures and likelihood interface

G1 predicts no generic freedom to drift G macroscopically. A visible drift obeys

$$\frac{\dot G}{G} \simeq -\frac{\dot\eta}{\eta}, \tag{43}$$

and must be carried by a conservation-closed residual or scalar-capacity sector. The three refined signature tests are:

1. **No free macroscopic G-drift.** Any variation of G is constrained by local-gravity, PPN, and propagation priors.
2. **Background–Weyl residual pattern.** Horizon occupancy residuals should produce a structured fingerprint $s < 3$, $\Sigma_0 < 0$, $\mu_0 \simeq 0$: background filling deviation and suppressed Weyl response through a common ledger variable, with the growth response near GR.
3. **Finite-area suppression.** Log or finite-area corrections give $\Delta G/G \sim -\alpha/(\eta_0 A)$ and are negligible on macroscopic screens unless an additional residual sector is activated.

A compressed likelihood interface has the schematic form

$$\chi^2_{\rm tot} = \chi^2_{\rm SN} + \chi^2_{\rm BAO} + \chi^2_{\rm RSD} + \chi^2_{E_G} + \chi^2_{\rm priors}. \tag{44}$$

Companion likelihood manuscripts compare ΛCDM, CPL, G1DE, and scalar-capacity variants. G1fit demonstrations have now been upgraded to a matched exact-pilot and medium-prior nested-evidence comparison (G1fit-real), using Pantheon+ full covariance, DESI DR2 BAO covariance, curated RSD, and compressed E_G. The six-model medium-prior evidence hierarchy is G1DE-M$_{3/4}$ $> M_\kappa >$ const-$\Sigma >$ G1DE2 $>$ CPL $>$ ΛCDM, with the projection-locked model ranking first. Top-control wide-prior sensitivity has been completed; full wide-prior baseline sensitivity and production evidence refinement remain pending.

14.1 Empirical status of the residual sector

The present status of the G1 residual sector is no longer purely schematic. A matched exact-likelihood and medium-prior nested-evidence comparison has been completed. The G1DE hierarchy uses an output-level memory response

$$\mathcal{R}_H(a) = R_0\widehat{\mathcal{R}}_H(a), \qquad \widehat{\mathcal{R}}_H(1) = 1, \qquad R_0 = 3 - s,$$

implemented through a Weyl-normalized transfer operator

$$\mathcal{W}_{\rm G1} = (1 - \kappa\mathcal{R}_H)\mathcal{W}_{\rm GR}.$$

This is an output normalization; it should not be confused with a unique microscopic source–kernel factorization $K_H * J_H$, which is treated in D6 as a deeper horizontal-lift layer.

$$\Sigma(a,k) - 1 = -\kappa R_0 \widehat{R}_H(a) + O(R_0^2), \qquad \mu(a,k) - 1 = O(R_0^2) + O(R_0\widehat{R}_H\mathcal{H}^2/k^2).$$

The projection-locked branch $\kappa = 3/4$, $\widehat{R}_H = X_b$ (with $X_b(1) = 1$ as the production-normalized output-response shape) gives $\Sigma(a) = 1 - \frac{3}{4}(3 - s)X_b(a)$, $\mu \simeq 1$. In the current medium-prior matched evidence pass, G1DE-M$_{3/4}$ ranks first among the tested six-model set. The detailed results and falsification ladder are in Companions D and E; the data paper is G1fit-real.

The refined empirical claim-status is therefore:

Table 3: Updated empirical status for the G1 residual sector.

Claim	Status
Finite screen entropy response	theoretical bridge
G1DE-0 background test	real-data full-covariance implemented
$s = 3$ ΛCDM-like limit	numerically validated
G1DE-2 exact pilot	survived exact full-likelihood pilot
$\Sigma_0 < 0$ Weyl response	pilot-level indication
$\mu_0 < 0$ growth response	not supported; $\mu_0 \simeq 0$
Medium-prior evidence completed; top-control wide-prior sensitivity completed; production validation pending	

A post-pilot projection-locked variant, G1DE-M$_{3/4}$, imposes $\mu(a,k) = 1$ (equivalently $\mu_0 = 0$) and $\Sigma(a) = 1 - \frac{3}{4}(3-s)X_b(a)$. In the current medium-prior matched evidence pass (G1fit-real), this four-parameter locked model ranks first among the tested six-model set; see Companions D, E for status. This is an empirical refinement of the residual sector, not a core bridge assumption.

The factorized production form is

$$\mathcal{R}_H(a) = R_0 \widehat{R}_H(a), \quad \widehat{R}_H(1) = 1, \quad R_0 = 3 - s,$$

implemented through a Weyl-normalized transfer operator $\mathcal{W}_{\mathrm{G1}} = (1 - \kappa\mathcal{R}_H)\mathcal{W}_{\mathrm{GR}}$. The active-filling shape $X_{\mathrm{fill}}(a)$ (peak-normalized) and the production-normalized memory shape $\widehat{R}_H(a) = X_{\mathrm{fill}}(a)/X_{\mathrm{fill}}(1)$ are related by a constant factor; the peak-normalized form is used in the internal model definition, while $\widehat{R}_H(1) = 1$ ensures identifiability of the projection coefficient κ and the residual amplitude R_0.

15 Horizon occupancy residuals

A minimal horizon occupancy residual model uses

$$\rho_H(a) = \chi_H(a)\rho_c(a), \qquad \rho_c(a) = \frac{3H^2(a)}{8\pi G}. \tag{45}$$

A lowest-order filling-closure law is

$$\frac{\mathrm{d}\chi_H}{\mathrm{d}\ln a} = s\chi_H\left(1 - \frac{\chi_H}{\chi_\infty}\right), \tag{46}$$

with solution

$$\chi_H(a) = \frac{\chi_\infty}{1 + Ba^{-s}}. \tag{47}$$

The effective equation of state is

$$w_H(a) = -1 - \frac{1}{3}\frac{\mathrm{d}\ln[\chi_H(a)H^2(a)]}{\mathrm{d}\ln a}. \tag{48}$$

The model is not intended as an arbitrary background fit. Its distinctiveness is a structured background–Weyl residual pattern

$$\chi_H(a) \Rightarrow H(a), \quad \Sigma(a), \quad \mu(a) \simeq 1, \quad \text{capacity closure}, \tag{49}$$

with the growth response near GR ($\mu_0 \simeq 0$) as a consistency condition rather than a required nonzero signal. This is more restrictive than a generic dark-energy or modified-gravity parametrization because the background and Weyl response share the same active-filling shape $\widehat{X}(a)$.

16 Holography and horizons

The holographic dictionary is naturally written in response-density language:

$$\frac{1}{4G_N} = \eta_{\text{screen}}. \tag{50}$$

For $\text{AdS}_3/\text{CFT}_2$,

$$c = \frac{3L}{2G_3}, \qquad \eta_{\text{AdS}_3} = \frac{1}{4G_3} = \frac{c}{6L}. \tag{51}$$

Thus central charge is interpreted as a boundary version of bulk screen capacity density. The Ryu-Takayanagi and FLM formulas are read as realizations of the screen-response principle:

$$S_R = \frac{A(\gamma_R)}{4G_N} + S_{\text{bulk}} + \cdots . \tag{52}$$

Similarly, black-hole and de Sitter horizons are exterior-accessible distinction ledgers with

$$S_H/k_{\text{B}} = \eta_{\text{BH}} A_H. \tag{53}$$

For Kerr-Newman horizons the first law

$$\mathrm{d}M = T_H \mathrm{d}S_H + \Omega_H \mathrm{d}J + \Phi_H \mathrm{d}Q \tag{54}$$

is read as a multi-channel capacity update law.

17 Failure registry

Table 4: Failure registry for FDS-G1 Core.

Failure condition	What fails	What may survive
No screen area extensivity	statistical core of G1a/G1b	other FDS ideas outside gravity
Wrong horizon coefficient	direct $G^{-1} \sim \eta$ calibration	screen statistics as entropy model
No local screen first law	thermodynamic gravity bridge	finite screen entropy response
Normal form not robust	G1micro hardening	response coupling and EFT sectors
No metric integrability	Einstein-like closure	nonmetric response theory
Unconstrained residuals	predictive value	core response formalism if residual taxonomy tightened
Scalar-capacity EFT ruled out	variable-capacity field model	microscopic/horizon-only corrections
No finite-index screen algebra net	D0 microscopic calibration	Core response framework as bridge
No boost modular calibration	$(2\pi/\hbar)$ modular route	standard horizon bridge
No metricization of anomaly-free sector	D0 Einstein derivation route	finite-access / residual framework
No passive horizon remnant	G1DE anomaly-remnant interpretation	phenomenological Weyl-transfer branch
G1DE loses to constant-Σ	output-shape $X_b(a)$	Weyl amplitude branch

Failure condition	What fails	What may survive
G1DE loses to free-κ	exact 3/4 isotropic compliance	weighted compliance branch
Data require $\lvert\mu-1\rvert \sim \lvert\Sigma-1\rvert$	Ward-suppressed Ricci leakage	finite-screen residual with density component
Free $A(a,k)$ required	Weyl normalization	generic dark-stress model, not G1DE
Production evidence loses to ΛCDM/CPL	G1DE observational branch	statistical and response core
Likelihood pipeline fails with real data	observational branch	theoretical bridge remains provisional

18 Companion manuscripts and archive structure

The recommended release structure is one flagship Core paper, six technical companions (A–F), one data-confrontation paper (G1fit-real), and the D0–D7 dark-sector closure sequence:

1. **Companion A: Finite Screen Statistical Mechanics.** Independent cells, interacting screens, and differential capacity.
2. **Companion B: Stochastic Screen Transitions.** Microtransition and Raychaudhuri-like capacity flow.
3. **Companion C: Scalar-Capacity EFT.** Horndeski and EFT-of-dark-energy embedding.
4. **Companion D: Likelihood and Falsification.** Predictions, observational interface, and compressed-data testing.
5. **Companion E: Horizon Occupancy Residuals.** Dark-sector ledger dynamics and model comparison.
6. **Companion F: Holography and Horizons.** Black-hole, de Sitter, and AdS/CFT dictionaries.
7. **G1DE-M$_{3/4}$: A Projection-Locked Background–Weyl Residual.** Matched exact likelihoods and medium-prior nested evidence for six late-time cosmology models.
8. **D0: Screen Modular Anomaly Inflow.** Einstein calibration of finite-access entropy density.
9. **D1: Finite-Access Sheaf Geometry.** Finite causal ledgers as a sheaf; gluing defects as residuals.
10. **D2: The 3/4 Projection Law.** Optical-stress closure and the projection coefficient.
11. **D3: Non-Equilibrium Screen Entropy.** Thermodynamic closure of the G1 residual.
12. **D4: Screen Influence Functional.** Unified influence framework for the dark-sector closure.
13. **D5: Weyl-Normalized Ward Completion.** Production likelihood interface and identifiability rules.
14. **D6: Maximum-Caliber Horizontal Lift.** Memory quotient, source–kernel identifiability, and maximum-caliber passive horizontal lift.
15. **D7: Finite Markov-Screen Realization.** A finite-state detailed-balance realization prototype for the D6 passive lift assumptions, showing how optical symmetry, a slow horizon mode, and Ward-stiff Ricci leakage realize the G1DE-M$_{3/4}$ normal form.

A code/data archive should include reproducible notebooks, demo compressed data, generated CSV files, figures, claim-status notes, prior-art boundaries, notation tables, and a failure registry.

19 Conclusion

FDS-G1 Core recasts the G1 program as entropy-response geometry. The central object is no longer a bare capacity density but a screen entropy functional $S_{\rm scr}[X]$ on a quotient state space. Its response one-form defines local area response; the area component controls local inverse coupling (stiffness) through Eq. (12), while $G_{\rm loc}$ is the corresponding coupling. The response Hessian controls stability and fluctuations; Onsager-type stochastic screen dynamics produces Raychaudhuri-like capacity-flow normal forms; all-null metric integrability and Ward/Bianchi closure generate Einstein-like equilibrium sectors; non-integrable response data appear as residuals.

The most compact statement is

$$\boxed{\text{Gravity-like geometry is the integrable hydrodynamic response of finite causal-screen entropy.}} \tag{55}$$

The reason finite screen entropy appears as the central object is that, in the FDS ontology, finite maintainable distinctions become physically accessible only through boundary ledgers; in gravity those ledgers are causal screens. The paper remains a bridge program. Its strength is not the claim to derive all of gravity from scratch, but the construction of a finite screen-capacity substrate, a local response principle, a normal-form dynamics, an EFT embedding, and a falsification interface that can be compared with existing gravitational, holographic, and cosmological frameworks. In the current medium-prior matched evidence pass reported by G1fit-real, the projection-locked residual model G1DE-$\mathrm{M}_{3/4}$ is preferred over free-κ, constant-Σ, free-response G1DE-2, CPL, and ΛCDM controls. This is a preliminary exact-pilot evidence result; top-control wide-prior sensitivity completed; full baseline sensitivity and production refinement remain necessary.

A Notation and response hierarchy

This appendix collects the main symbols. The purpose is to keep the response hierarchy explicit:

$$S_{\rm scr}[X] \longrightarrow \omega = \eta_i \, \mathrm{d}X^i \longrightarrow \eta_{\rm loc} = \omega(\delta_A) \longrightarrow G_{\rm loc} = \frac{c^3}{4\hbar}[\eta_{\rm loc}]^{-1}.$$

The average density $\bar{\eta}(A)$, the thermodynamic density $\eta_{\rm eff}$, and the local response density $\eta_{\rm loc}$ are related but not interchangeable.

Table 5: Notation table for FDS-G1 Core.

Symbol	Meaning	Role
$S_{\rm scr}[X]$	screen entropy functional on quotient state space	primitive object
$\mathcal{X}_{\rm raw}$	raw screen variables including gauge labels	pre-quotient bookkeeping
$\mathcal{X}_{\rm phys}$	physical quotient state space $\mathcal{X}_{\rm raw}/\mathcal{G}_{\rm scr}$	domain of $S_{\rm scr}$
$A(x)$	local area element or cell-count density	area direction
q_{AB}	screen shape metric	shape/shear sector
χ	occupancy or horizon residual fraction	dark/residual variable
Φ	flux load crossing screen	focusing source proxy
M	memory or hidden response mode	non-Markovian sector
ω	response one-form $\eta_i \mathrm{d}X^i$	differential response object
$\eta_{\rm loc}$	area component of ω	local inverse-coupling response density
$\eta_\delta(A)$	homogeneous differential density	finite-area response
$\eta_{\rm eff}$	thermodynamic area-law density	interacting screen limit

Symbol	Meaning	Role
G_{loc}	local effective coupling under horizon bridge; $G_{\mathrm{loc}}^{-1} \propto \eta_{\mathrm{loc}}$	inverse response stiffness
$\mathcal{H}_{ij}$	entropy response Hessian	stability/fluctuation metric
θ	logarithmic screen expansion	capacity-flow variable
σ_{AB}	traceless shape flow	shear-like dissipation
Ξ_{res}	residual forcing in capacity flow	nonintegrable/stochastic sector
$\mathcal{R}_{\mu\nu}$	covariant residual representative	metric closure defect
φ	scalar-capacity field $\ln(\eta/\eta_0)$	EFT slow mode

B Detailed claim map: derived, bridged, and modeled

The word "derivation" is local to a specified model. A screen-cell entropy formula may be derived in a toy model, while an Einstein-like equation remains a conditional bridge unless all geometric inputs are derived internally.

Table 6: Detailed claim map.

Statement	Status	Reason
$S_A/k_{\mathrm{B}} = (A/a_0)\ln m$	toy derivation	follows from independent finite cells
$S_A/k_{\mathrm{B}} = \eta_{\mathrm{eff}} A + o(A)$	statistical universality claim	requires screen-local finite-state noncritical assumptions
$\eta_{\mathrm{loc}} = k_{\mathrm{B}}^{-1}\delta S/\delta A$	definition	area component of response one-form
$G_{\mathrm{loc}} = (c^3/4\hbar)[\eta_{\mathrm{loc}}]^{-1}$	conditional bridge	requires horizon entropy normalization
$\dot{\theta} = -a\theta^2 - b\sigma^2 - c\Phi + \Xi$	normal-form derivation	follows from low-order stochastic transition assumptions
$\Phi \leftrightarrow R_{\mu\nu}k^\mu k^\nu$	geometry bridge	standard local-screen/Raychaudhuri identification
Einstein-like closure	conditional bridge	needs all-null integrability and Ward/Bianchi conservation
$\varphi = \ln(\eta/\eta_0)$ scalar EFT	model	conservative scalar-tensor realization of capacity mode
G1DE occupancy residual	phenomenological model	tests correlated background/growth/lensing deviations
Holographic dictionary	compatibility bridge	interprets $1/(4G_N)$ and central charge as response density

C Expanded entropy-response toy model

The simple two-variable model illustrates how finite-area corrections, occupancy residuals, Hessian stability, and apparent nonintegrability fit into one framework. Let

$$s(A,\chi) = \eta_0 A + \alpha \ln(A/A_0) + \beta A\chi - \frac{\lambda}{2}\chi^2. \tag{56}$$

The area and occupancy responses are

$$\eta_A = \eta_0 + \frac{\alpha}{A} + \beta\chi, \qquad \eta_\chi = \beta A - \lambda\chi. \tag{57}$$

The homogeneous local coupling is

$$G_A(A,\chi) = \frac{c^3}{4\hbar}\left[\eta_0 + \frac{\alpha}{A} + \beta\chi\right]^{-1}. \tag{58}$$

Thus finite-area corrections and occupancy residuals enter the same response coefficient.

The Hessian is

$$\mathcal{H} = -\begin{pmatrix} \partial_A^2 s & \partial_A\partial_\chi s \\ \partial_\chi\partial_A s & \partial_\chi^2 s \end{pmatrix} = \begin{pmatrix} \alpha/A^2 & -\beta \\ -\beta & \lambda \end{pmatrix}. \tag{59}$$

Local stability in this sector requires

$$\alpha/A^2 > 0, \qquad \frac{\alpha\lambda}{A^2} - \beta^2 > 0. \tag{60}$$

If the area-occupancy coupling is too strong, the response Hessian loses positivity. In a physical model this signals instability, a missing stabilizing variable, or the need for a different coarse-grained description.

Now add a hidden memory variable

$$s(A,\chi,M) = \eta_0 A + \alpha\ln(A/A_0) + \beta A\chi + \gamma AM - \frac{\lambda}{2}\chi^2 - \frac{\mu}{2}M^2. \tag{61}$$

On (A,χ,M) the response one-form is exact. If the observed coarse variables are only (A,χ) but M follows a history-dependent relation $M = M[A,\chi;\mathrm{path}]$, the reduced response is generally not exact. The apparent curl is then a diagnostic of projection, memory, or genuine residual physics. This is the simplest mathematical model for the epistemic/physical residual distinction.

D Three levels of integrability

The term "integrability" appears in three distinct senses.

Level I: entropy-potential integrability. This is the condition

$$\omega = \mathrm{d}s_{\mathrm{scr}}, \qquad \mathrm{d}\omega = 0. \tag{62}$$

It asks whether the chosen screen response variables are sufficient to define an entropy potential. Failure at this level often indicates missing coarse variables or memory.

Level II: metric all-null integrability. This asks whether local null response data can be represented by a symmetric tensor field,

$$\mathcal{F}_p(k) = E_{\mu\nu}(p)k^\mu k^\nu. \tag{63}$$

All-null data determines only the trace-free part, because $g_{\mu\nu}k^\mu k^\nu = 0$. The undetermined trace becomes an integration freedom, represented in Einstein gravity by the cosmological constant sector.

Level III: Ward/Bianchi dynamical integrability. This asks whether the tensor envelope satisfies a covariant conservation identity,

$$\nabla^\mu E_{\mu\nu} = 0. \tag{64}$$

Residual sectors are admissible only if the total effective source is conserved,

$$\nabla^\mu(T_{\mu\nu} + \mathcal{R}_{\mu\nu}) = 0. \tag{65}$$

These three notions should not be conflated. A closed entropy response one-form does not by itself prove metric geometry; metric all-null reconstruction does not by itself prove dynamics; and Bianchi closure is a further gauge/Ward condition.

E Open problems

FDS-G1 Core leaves three major scientific tasks:

1. **Coefficient problem.** Explain, rather than merely bridge to, the Bekenstein-Hawking normalization $1/(4\ell_P^2)$.
2. **Microgeometry problem.** Strengthen the normal-form theorem into a broader derivation of focusing coefficients and geometric identification.
3. **Real-data confrontation.** A matched exact-likelihood and medium-prior nested-evidence comparison (G1fit-real) has been executed; the remaining tasks are full wide-prior baseline sensitivity, production ($d \log Z$) refinement, expanded weak-lensing/E_G likelihoods, and independent pipeline replication to test the background–Weyl residual pattern at survey quality.

These are not defects in the stated bridge program. They are the natural next steps if the program is to be promoted from finite-capacity bridge to a candidate microscopic account of gravity-like geometry.

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